Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 5
Subject: TCD: Junior Designer Challenge	**Date: Week 10 (2 Nov to 6 Nov)**
Class Description: Grade 5 TCD class, Junior Designer Challenge Session 2 of 5 (1 hour 30 minutes). Students study an existing car logo from the teacher's reference sheet, break it into simple shapes and rebuild it as an accurate 3D model in Tinkercad.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	3D Modelling a Car Logo	
Standard(s)	Math: 5.G.B.3 and 5.G.B.4 (classify shapes by properties); Visual Arts: analyse form and symmetry; Technology: CAD modelling	
Learning Objective(s) (WALT): We are learning to break a logo into simple shapes and rebuild it as a 3D model.		**Success Criteria (WILF):** I can find the basic shapes inside a car logo. I can use rotate, duplicate and mirror to build symmetrical parts. I can use holes to cut out inner shapes. I can scale my model to 60 mm and give it a 4 mm base.
Literacy Strategies	Shape-spotting sheet with the words circle, ring, star, triangle, oval and letter; sentence frame: This logo is made of ...	
Assessment	Shape breakdown sheet; Tinkercad model checked against the reference; live quiz on the TCD website.	
Targeted Vocabulary	logo, symmetry, mirror, duplicate, rotate, scale, ring, reference	
Skills	Visual analysis, geometry, digital modelling, accuracy	
National Identity Integration	Discuss the UAE's car culture, from the Dubai Police supercar fleet to the Etihad Rail and electric vehicles, and how strong logos	

	help companies become known around the world.
AI/Digital Learning	Tinkercad; printed logo reference sheet (teacher prepared); hook video; live quiz on the TCD website. Presentation: Grade5_Week10_Presentation.pptx. Worksheet: Grade5_Week10_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Logo Detective (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (20 minutes)	Model breaking a logo into shapes (slide 6). Model a ring cut-out and overlapping rings (slide 7). Model duplicate, rotate and mirror (slide 8) and scaling (slide 9).
We Do (15 minutes)	Class breaks down one logo together on the board. Students build one ring together, step by step (slide 10).
You Do (35 minutes)	Students model their chosen logo (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Choose a logo made of rings or circles and follow the ring step card.** **On: Choose a logo with a star, triangle or letter and use rotate or mirror.** **Above: Choose a complex logo, add a raised border and explain how you used symmetry.**
Closure/ Reflection (10 minutes)	Side-by-side check: model next to the reference. One fix for next week. Exit ticket (slide 15). Save and name the file Grade5_CarLogo_YourName.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: shape word cards and sentence frame. SEN: ring-based logo with a step card; buddy support. G&T: complex logo with border; explain the symmetry used.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☑ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☐ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☑ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☐ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: